

MLCommons Medical AI Working Group

Response to HHS Request for Information and Proposed Rule

Advancing the Quintuple Aim Through AI-Enabled Healthcare

Response to:

1. Request for Information: Accelerating the Adoption and Use of Artificial Intelligence as Part of Clinical Care (Document No. 2025-23641, RIN 0955-AA13)
2. Proposed Rule: Health Data, Technology, and Interoperability: ASTP/ONC Deregulatory Actions To Unleash Prosperity (Document No. 2025-23896, RIN 0955-AA09)

Submitting Organization

MLCommons' Medical AI Working Group

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About MLCommons and the Medical AI Working Group

MLCommons is a non-profit consortium that aims to accelerate the benefits of machine learning and artificial intelligence. Our members and partners include over 125 organizations from around the world, many of which are leading technology companies, startups, academics, and nonprofits that are actively researching, developing, and deploying artificial intelligence products for customers. Critically, our founding membership includes academic researchers at the forefront of machine learning research, and the research community continues to be core to our membership helping to lead many of our working groups. MLCommons acts as a neutral nexus for commercial and non-commercial actors to collaborate on tools that advance the field.

We create, operate and maintain community assets, especially benchmarks and datasets, that facilitate developing and evaluating artificial intelligence (AI) systems in pursuit of our mission to “make artificial intelligence better for everyone.”^[1] The original project that brought MLCommons into being is a benchmarking suite called MLPerf, which provides objective evaluations of training and inference speed for AI hardware and software.^[2] These measurements enable a fair comparison of competing systems, accelerate ML progress through fair and useful measurement, enforce reproducibility to ensure reliable results, and do so in an open and collaborative way to keep benchmarking affordable for all participants.

The MLCommons Medical AI Working Group develops benchmarks and best practices that help accelerate AI development in healthcare through an open, neutral, and scientific approach. Our efforts are geared towards creating the fabric necessary for proper benchmarking of medical AI through:

- (1) developing the federated MedPerf platform^[3] and GaNDLF framework^[4],
- (2) establishing shared standards for benchmarking medical AI,
- (3) incorporating and disseminating best practices, and
- (4) creating a robust medical AI ecosystem by partnering with key global stakeholders across sectors.

Our Platform & Tools:

Tool	Purpose	Relevance to HHS Initiatives
MedPerf	Federated benchmarking platform for evaluating AI on real-world medical data without data leaving provider premises	Enables AI evaluation across varied populations while preserving privacy; supports post-market monitoring
GaNDLF	Zero/low-code framework for developing ML workflows in healthcare	Accelerates AI development with standardized, reproducible pipelines

Croissant	Metadata format for ML-ready datasets[5]	Addresses data interoperability and reproducibility challenges
AILuminate	Benchmark suite for evaluating AI risk and reliability[6]	Supports safety evaluation for clinical AI systems

MedPerf is currently being utilized in the **largest federated AI research study for postoperative glioblastoma patients** (FeTS 2.0), demonstrating that federated evaluation can operate at scale across 53 institutions globally.[7] This study analyzed 10,208 follow-up timepoints from postoperative glioblastoma patients, achieving record performance that exceeded centralized approaches while ensuring patient imaging data never left institutional premises.

Our working group has included participation from patient groups, academic medical centers (e.g., Harvard Medical School, Indiana University, Stanford University School of Medicine, University of Pennsylvania, Weill Cornell Medicine, Dana-Farber Cancer Institute, among others), commercial companies (e.g., Google, NVIDIA, IBM Research, Intel, Hugging Face, Microsoft, John Snow Labs, etc.), healthcare institutions (e.g., Brigham and Women’s Hospital, Dana-Farber Cancer Institute, etc.), and non-profit organizations spanning multiple countries.

For more information, visit mlcommons.org/medical or contact medical-chairs@mlcommons.org

Executive Summary

This response addresses two interconnected HHS initiatives: the Request for Information on Accelerating AI Adoption in Clinical Care (RIN 0955-AA13) and the proposed deregulatory rule on Health Data, Technology, and Interoperability (RIN 0955-AA09). We support both initiatives and believe they represent a historic opportunity to advance the **Quintuple Aim** through responsible AI deployment:

Quintuple Aim Dimension	How AI Can Advance It	How These HHS Initiatives Enable It
Improving Population Health	AI-enabled screening, early detection, risk stratification across populations	FHIR APIs enable population-level analytics; reduced information blocking enables public health AI
Enhancing Patient Experience	Reduced wait times, better care coordination, personalized treatment	Interoperability enables AI to synthesize information across providers
Reducing Per Capita Cost	Automation of administrative tasks, reduced unnecessary testing, optimized resource allocation	Deregulation reduces compliance burden; federated evaluation reduces duplicate validation costs; transparency reduces friction
Improving Provider Work Life	Clinical documentation assistance, decision support, reduced alert fatigue	Streamlined AI integration through standardized APIs; reduced administrative overhead
Healthcare Delivery Optimization	Performance monitoring across demographics, identification of performance variations, expanded access	Federated evaluation across geographically varied sites; standardized data enables performance optimization

Our recommendations are drawn from our collective expertise and experiences as a technical collaborative community of ML researchers and engineers in industry and academia. We have previously provided detailed technical recommendations to the FDA regarding real-world evaluation of AI-enabled medical devices (Docket No. FDA-2025-N-4203), and this response builds upon that foundation while addressing the broader policy questions raised by HHS.

AI is poised to transform healthcare delivery, but adoption can be slow unless standardized evaluation of available AI tools matures to the point of establishing trust and credibility amongst adopters and patients alike. MLCommons is the leading AI evaluation organization in the world, and its Medical working group is an open collaborative community working to bring multidisciplinary expertise in AI evaluation to the problems of healthcare delivery specifically through a federated evaluation approach that respects data and model ownerships.

Key Themes of Our Response:

1. **Interoperability is foundational for AI and the Quintuple Aim:** The proposed FHIR-forward approach directly enables AI evaluation, benchmarking, and deployment at scale. Standardized data access through FHIR APIs creates the technical infrastructure necessary for AI systems to integrate into clinical workflows and deliver on all five aims. AI governance described through technical policies can bring transparency and integrity of sensitive workloads.
2. **Federated evaluation addresses critical barriers while optimizing delivery:** Many barriers to AI adoption stem from data access challenges and evaluation difficulties. Our MedPerf platform demonstrates that federated approaches can enable robust AI evaluation while preserving data privacy, security, and enabling performance assessment across varied populations.
3. **Open, neutral benchmarking accelerates trust and reduces costs:** Industry-driven, transparent evaluation methods provide the foundation for clinician and patient trust in AI systems while reducing duplicative validation efforts. HHS can support these efforts through cooperative agreements and recognition of community-developed standards.
4. **Deregulation should preserve AI safety guardrails:** While we support reducing unnecessary certification burden, we encourage HHS to ensure that mechanisms for evaluating AI safety and effectiveness remain robust, particularly for high-risk clinical applications. Technical policies can help facilitate such requirements.
5. **American leadership requires coordinated action:** The United States can cement its position as the global leader in clinical AI through the coordinated approach represented by these initiatives. Open standards and federated infrastructure developed here can become the global model.

Part I: Response to RFI on Accelerating AI Adoption in Clinical Care

Framing: AI as an Enabler of the Quintuple Aim

Before addressing specific questions, we emphasize that AI adoption should be evaluated through the lens of the Quintuple Aim. AI tools that improve one dimension while degrading another (e.g., improving efficiency while creating performance gaps across populations) should be considered unfit for adoption in clinical workflow. The evaluation and monitoring frameworks we propose below are designed to assess AI impact across all five dimensions.

Regulation

HHS seeks feedback on how current regulations impact AI adoption. We offer the following observations:

Alignment Across Agencies is Critical

The healthcare AI ecosystem involves multiple regulatory touchpoints: FDA for medical devices, CMS for reimbursement, and ONC for health IT certification. We commend HHS for taking a coordinated approach through this RFI. However, we note that AI systems often span multiple regulatory categories. For example, an AI system that assists with clinical decision-making may:

- Require FDA clearance as a medical device
- Need to integrate with certified EHR systems under ONC rules
- Affect billing and documentation practices under CMS programs

We recommend that HHS establish clear guidance on regulatory pathways that accounts for these overlapping jurisdictions. A unified framework would reduce uncertainty for developers, accelerate time-to-deployment for beneficial AI tools, and ultimately advance the Quintuple Aim by removing barriers to innovation.

Support for Non-Device AI Evaluation

For AI tools that fall outside FDA medical device jurisdiction, there is currently limited guidance on appropriate evaluation methods. Our working group has developed extensive experience in AI evaluation frameworks that could inform voluntary guidance for non-device clinical AI.

We propose the following evaluation framework aligned with the Quintuple Aim:

Evaluation Domain	Key Metrics	Quintuple Aim Alignment
Clinical Safety	False negative rates, adverse events, time-to-intervention delays	Population Health, Patient Experience
Effectiveness	Sensitivity/specificity, diagnostic concordance, outcome prediction accuracy	Population Health
Reliability	System uptime, prediction confidence, error rates by scenario	Patient Experience, Provider Work Life

Performance Consistency	Performance stratified by demographics, geographic access, language support	Healthcare Delivery Optimization
Efficiency	Time savings, cost reduction, workflow integration	Per Capita Cost, Provider Work Life
Provider Impact	Override rates, alert fatigue indicators, documentation burden	Provider Work Life

Reimbursement

Payment policy significantly influences AI adoption and directly affects whether AI advances or undermines the Quintuple Aim.

Evidence Generation Support

Many promising AI tools lack the real-world evidence necessary to support coverage decisions. HHS could accelerate evidence generation by:

1. Supporting federated evaluation networks that enable AI assessment across varied clinical sites without requiring data centralization
2. Funding prospective studies of AI tool impact on clinical outcomes and costs
3. Establishing data sharing agreements that facilitate AI evaluation while protecting patient privacy
4. Requiring AI tools to demonstrate performance across varied populations as a condition of coverage (optimizing healthcare delivery)

Research & Development

HHS's research ecosystem can catalyze AI innovation that advances the Quintuple Aim.

Public-Private Partnerships for Benchmarking Infrastructure

MLCommons operates as a neutral convener of industry, academic, and non-profit stakeholders to develop benchmarking tools. This model has proven effective for AI hardware/software benchmarking through MLPerf and is now being applied to medical AI through MedPerf. HHS could support similar efforts through:

- Cooperative Research and Development Agreements (CRADAs) with benchmark developers
- Funding for broad, representative benchmark datasets that enable performance optimization
- Recognition of community-developed evaluation standards (such as MLCommons' Croissant for dataset metadata and MLCommons' AILuminate for AI safety)
- Support for federated infrastructure that enables evaluation without data centralization

Federated Research Infrastructure: The FeTS 2.0 Model

Our MedPerf platform demonstrates that federated evaluation can enable AI assessment on real-world clinical data without requiring data centralization. The **Federated Tumor Segmentation (FeTS) 2.0 study** provides concrete evidence of this approach at scale:

Metric	FeTS 2.0 (Federated)	BraTS 2024 (Centralized)
Participating institutions	53 global sites	Centralized
Training data	8,100 timepoints	1,350 timepoints
Validation data	2,108 timepoints	397 timepoints
Total patient data	10,208 timepoints	1,747 timepoints
Peritumoral edema DSC	0.95	0.89
Enhancing tumor DSC	0.94	0.79
Resection cavity DSC	0.89	0.78
Necrosis DSC	0.77	0.81

Key findings from FeTS 2.0:

Scale without data sharing: 53 institutions contributed data while patient imaging never left institutional premises. Only trained model weights and validation scores were shared between collaborators.

Superior performance: The federated approach achieved record Dice Similarity Coefficients, outperforming the centralized BraTS 2024 study in three of four segmentation classes (peritumoral edema, enhancing tumor, resection cavity) while achieving comparable results for necrosis.

Privacy preservation: Executed through MLCommons' MedPerf orchestrator and Intel's OpenFL framework, demonstrating that privacy-preserving approaches do not sacrifice performance.

Clinical impact: The resulting AI model enables reliable delineation of postoperative regions-of-interest, supporting standardized glioblastoma response assessment and adaptive therapeutic decision-making worldwide.

This study demonstrates that **federated approaches can achieve superior results compared to centralized methods** while respecting data privacy, a finding directly relevant to HHS policy on data access and AI evaluation.

We recommend that NIH and other HHS research entities invest in federated infrastructure that can support both AI development and post-deployment monitoring while enabling performance optimization across varied populations.

Responses to Specific Questions

Question 1: What are the biggest barriers to private sector innovation in AI for health care and its adoption and use in clinical care?

Based on our experience working with a broad range of stakeholders, the primary barriers include:

Barrier	Description	Impact on Quintuple Aim	Potential Solution
Data Access	Privacy regulations, institutional policies, and technical barriers limit access to clinical data	Limits AI development and performance optimization	Technical Policies, Metadata (Croissant) Federated evaluation (MedPerf)
Evaluation Infrastructure	Developers lack access to representative populations for prospective evaluation	Undermines safety and performance consistency	Standardized benchmark datasets; federated testing
Regulatory Uncertainty	Limited guidance for non-device AI	Delays deployment of beneficial tools	Harmonized HHS guidance
Integration Complexity	Fragmented EHR landscape requires custom integrations	Increases cost, limits adoption	FHIR API standardization
Trust Deficit	Clinicians lack confidence in AI performance	Reduces adoption of effective tools	Open benchmarking; transparency requirements
Performance Variation Concerns	Fear that AI will create performance gaps across populations	Limits adoption in resource-limited settings	Stratified performance reporting

Question 2: What regulatory, payment policy, or programmatic design changes should HHS prioritize?

We recommend HHS prioritize changes that advance all five dimensions of the Quintuple Aim:

1. **Harmonized guidance on non-device clinical AI evaluation:** Clear standards for evaluation methods, performance metrics, and ongoing monitoring for AI tools outside FDA jurisdiction. This guidance should explicitly require performance optimization assessment.
2. **Support for federated evaluation infrastructure:** Funding and policy support for platforms that enable AI evaluation on real-world data while preserving privacy and enabling performance assessment across varied populations.

3. **Accelerated coverage pathways for AI tools with demonstrated Quintuple Aim impact:** Streamlined processes for incorporating AI tools that demonstrate improvement across multiple aim dimensions.
4. **Interoperability requirements that support AI integration:** The proposed FHIR-forward approach in the deregulatory rule directly addresses this need.
5. **Performance optimization requirements:** AI tools seeking coverage should demonstrate consistent performance across demographic groups, with particular attention to populations underrepresented in training data.

Specific regulatory provisions that could be revisited include:

- **45 CFR 164.514 (HIPAA de-identification):** Clarify that federated evaluation approaches where protected health information never leaves covered entities can satisfy HIPAA requirements. This would unlock evaluation across varied populations without compromising privacy.
- **42 CFR 414 (Medicare Part B):** Consider demonstration projects for AI tool reimbursement in value-based arrangements, with explicit requirements for performance optimization assessment.
- **42 CFR 425 (MSSP):** Allow ACOs to count AI-enabled quality improvement activities toward shared savings calculations.

Question 3: For non-medical devices, what novel legal and implementation issues exist?

Key issues include:

1. **Algorithm transparency:** Requirements for disclosing AI system characteristics, limitations, and performance vary. Standardized transparency requirements would benefit all stakeholders. MLCommons has developed approaches for AI documentation through model cards and our ALLuminate framework that could inform such requirements.
2. **Continuous learning systems:** AI systems that update based on local data raise questions about validation requirements and version control. Our working group has developed approaches for versioning and lineage tracking that could inform guidance.
3. **Performance optimization and calibration:** AI systems may perpetuate or amplify existing differential healthcare outcomes. Emerging research has identified specific systematic error recognition and correction strategies, but legal frameworks for addressing algorithmic performance variation in healthcare are underdeveloped. HHS could provide guidance on acceptable performance differentials across demographic groups.
4. **Data ownership and access:** When AI systems are trained or fine-tuned on institutional data, questions arise about data ownership, model ownership, and ongoing access rights. Information blocking regulations should address these scenarios.

Question 4: For non-medical devices, what are the most promising AI evaluation methods?

Based on our experience developing and deploying evaluation infrastructure, we recommend a comprehensive framework:

Pre-Deployment Evaluation Framework:

Evaluation Type	Methods	Tools/Standards
Performance Assessment	Benchmark testing on standardized datasets[8]	MedPerf, domain-specific benchmarks
Performance Optimization Assessment	Stratified performance analysis across demographics	Croissant metadata for demographic tracking
Robustness Testing	Evaluation against data quality variations, distribution shifts	Statistical tests (MMD, KS)
Safety Evaluation	Failure mode analysis, edge case testing	ALLuminate, curated edge case repositories
Human Factors	Usability testing, workflow integration assessment	FDA human factors guidance (adapted)

Post-Deployment Monitoring Framework:

Monitoring Level	Frequency	Activities	Quintuple Aim Focus
Continuous	Real-time	System output tracking, safety signals	Patient safety
Weekly	7 days	Automated performance summaries, statistical process control	Reliability

Monthly	30 days	Comprehensive reports with trend analysis, optimization metrics	Healthcare Delivery Optimization, effectiveness
Quarterly	90 days	Clinical expert review of edge cases and failure modes	Safety, provider experience
Annual	Yearly	Full re-evaluation against updated standards	All dimensions

Statistical Methods for Drift Detection:

- Control charts (CUSUM, EWMA) for detecting performance drift
- Maximum mean discrepancy (MMD) for distribution shift detection
- Sequential probability ratio tests for early warning
- Calibration assessment comparing predictions to outcomes

HHS could support these methods through:

- Cooperative agreements with organizations developing evaluation infrastructure
- Grant funding for evaluation tool development and validation studies
- Prize competitions for innovative approaches to AI monitoring
- Recognition of community-developed standards as acceptable evaluation methods

Question 5: How can HHS best support private sector activities to promote innovative and effective AI use?

We recommend actions that reduce barriers while ensuring AI advances the Quintuple Aim:

1. **Recognition of industry-developed standards:** HHS could formally recognize community-developed evaluation standards (like those from MLCommons) as acceptable methods for demonstrating AI safety and effectiveness. This would reduce duplicative validation costs.
2. **Public benchmarking infrastructure:** Support for open, neutral benchmarking platforms that enable fair comparison of AI systems. This promotes competition and drives quality improvement.
3. **Federated evaluation networks:** Fund the development of federated infrastructure that allows AI evaluation across varied clinical sites without data centralization. This directly enables performance optimization.
4. **Accreditation pathway development:** Work with professional societies and accreditation bodies to develop AI competency frameworks for clinical institutions.

5. **Safe harbor for good-faith evaluation:** Provide legal protection for institutions that share data for AI evaluation purposes and for developers that transparently report performance issues.
6. **Workforce development:** Support training programs for clinical informaticists, data scientists, and AI safety specialists who can oversee AI deployment.

Question 6: Where have AI tools deployed in clinical care met or exceeded performance expectations and where have they fallen short?

Our working group has observed the following patterns:

Successes (advancing multiple Quintuple Aim dimensions):

Application	Quintuple Aim Impact	Key Success Factors
Diabetic retinopathy screening	Population health, expanded access, cost	Well-defined task, clear ground truth, regulatory clarity
Clinical documentation/ambient scribing	Provider work life, cost	Addresses acute pain point, measurable time savings
Sepsis early warning	Population health, patient experience	Clear clinical value, integration into existing workflows
Radiology triage (critical findings)	Patient experience, population health	Reduces delays for urgent cases

Challenges (where AI has fallen short):

Challenge	Quintuple Aim Impact	Root Cause
Performance degradation on new populations	Healthcare Delivery Optimization	Training data not representative; lack of federated evaluation
Alert fatigue	Provider work life	High false-positive rates; poor workflow integration
Outcome improvement unclear	Population health, cost	Intermediate metrics don't translate to patient outcomes
Variable performance across demographics	Healthcare Delivery Optimization	Insufficient evaluation across subgroups

Integration friction	Cost, provider work life	Lack of standardized APIs; custom integration required
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Greatest Potential (pending barrier removal):

- AI systems that reduce administrative burden (advancing provider work life and cost)
- Population health management and risk stratification (advancing population health and delivery optimization)
- Care coordination across fragmented health systems (advancing patient experience)
- Federated AI that learns from distributed data while preserving privacy (advancing all dimensions)

Question 7: Which roles within health care organizations have the most influence on AI adoption?

Based on our interactions with a broad range of health systems:

Role	Influence Level	Primary Concerns	How HHS Can Help
CMIO/CMAO	Very High	Clinical safety, workflow integration, evidence	Clear evaluation standards, recognized benchmarks
CIO/CTO	High	Security, integration, vendor management	FHIR standardization, security guidance
Clinical Champions	High	Specialty-specific value, peer acceptance	Specialty society engagement, pilot funding
Legal/Compliance	High (veto power)	Liability, regulatory risk	Clear guidance, safe harbors
CFO/Finance	High	ROI, reimbursement	Coverage pathways, value demonstration
Nursing Leadership	Medium-High	Workflow, patient safety, staff burden	Workflow integration requirements
Quality/Safety Officers	Medium	Outcomes, adverse events	Monitoring standards, reporting requirements

Primary administrative hurdles include:

- Lengthy security and privacy review processes (6-18 months typical)
- Lack of standardized evaluation criteria leading to duplicative reviews
- Integration complexity with existing systems
- Unclear governance structures for AI oversight
- Procurement processes not designed for AI (SaaS, continuous updates)

Question 8: Where would enhanced interoperability widen market opportunities and accelerate AI development?

Enhanced interoperability would have the greatest impact in areas that advance the Quintuple Aim:

Priority Data Types for AI:

Data Type	Current State	AI Opportunity	Quintuple Aim Impact
Clinical notes	Limited standardization	NLP for documentation, clinical insights	Provider work life, population health
Medical imaging	DICOM standard but metadata inconsistent	Diagnostic AI, treatment planning	Population health, cost
Patient-generated data	Fragmented, non-standard	Remote monitoring, chronic disease management	Population health, patient experience
Social determinants	Rarely structured	Risk stratification, care coordination	Healthcare Delivery Optimization, population health
Prior authorizations	Manual, fax-based	Automation, real-time decisions	Cost, provider work life
Quality measures	Retrospective, burdensome	Real-time quality monitoring	Population health, cost

Data Standards Priorities:

- FHIR-based APIs for consistent data access (strongly support proposed rule)
- Standardized data quality indicators (enables AI to assess input reliability)
- Croissant-compatible metadata for ML datasets (enables reproducibility)
- Common representation of temporal relationships (critical for predictive AI)
- Standardized demographic data elements (enables performance optimization assessment)

Benchmarking Tools Needed:

- Reference implementations for AI evaluation on FHIR data
- Synthetic data generation tools that preserve statistical properties
- Federated evaluation infrastructure compatible with certified EHR APIs
- Performance optimization frameworks with standardized demographic categories

The proposed FHIR-forward approach in the deregulatory rule directly addresses many of these needs and should be finalized.

Question 9: What challenges do patients and caregivers wish to see addressed by AI?

Based on input from patient representatives in our working group:

Desired AI Applications (mapped to Quintuple Aim):

Patient/Caregiver Need	AI Solution	Quintuple Aim Dimension
Faster diagnosis	Triage AI, diagnostic support	Patient Experience
Care coordination	Information synthesis across providers	Patient Experience, Cost
Understanding medical information	Plain language explanations, health literacy tools	Patient Experience, Healthcare Delivery Optimization
Medication safety	Interaction checking, adherence support	Population Health, Patient Experience
Caregiver support	Care planning tools, resource navigation	Patient Experience, Provider Work Life
Access in rural and resource-limited areas	Telehealth AI, remote diagnostics	Healthcare Delivery Optimization
Reduced out-of-pocket costs	Administrative automation, prior auth	Cost

Patient Concerns:

Concern	Mitigation Strategy	HHS Role
Privacy/security	Federated approaches, strong encryption	Clarify HIPAA applicability
Transparency	Disclosure when AI involved in care	Require transparency standards
Performance variations	Stratified performance monitoring	Require performance optimization assessment

Human oversight	Maintain physician role	Guidance on appropriate autonomy levels
Broad access	Ensure AI benefits reach all populations	Coverage requirements, rural health focus

Question 10: Are there specific areas of AI research that HHS should prioritize?

We recommend prioritizing research that advances the Quintuple Aim:

Research Area	Quintuple Aim Impact	Current Gap	Recommended Investment
Evaluation methodology	All dimensions	Limited real-world assessment methods	Federated evaluation infrastructure
Performance consistency	Healthcare Delivery Optimization	Systematic error detection and correction immature	Broad benchmark datasets, optimization metrics
Human-AI interaction	Provider work life, safety	Limited understanding of optimal collaboration	Clinical workflow studies
Continuous learning	All dimensions	Safety of adaptive systems unclear	Methods for safe model updates
Multimodal AI	Population health, effectiveness	Integration of imaging, notes, labs, genomics	Interoperability standards for AI
Federated learning	Healthcare Delivery Optimization, privacy	Privacy-preserving training at scale	Infrastructure and methods research
Administrative AI	Cost, provider work life	Automation of prior auth, documentation	Workflow automation studies

10a. Published findings on adopted AI tools:

The literature on deployed clinical AI is growing but remains limited. Key sources include: - FDA's database of authorized AI/ML-enabled medical devices (1,000+ devices)[9] - Published post-market studies (relatively rare; systematic selection toward positive results) - Health system implementation reports (often not peer-reviewed) - Industry-sponsored studies (potential conflicts of interest)

We recommend HHS support systematic evidence generation for deployed AI tools through: - Real-world evidence initiatives with standardized outcome measures - Registry requirements for AI-enabled devices - Federated post-market surveillance networks

10b. Literature on costs, benefits, and transfers:

The literature on AI economics in healthcare is nascent.[10] Most studies focus on potential benefits (improved accuracy, time savings) rather than measured outcomes. Key gaps include:

- Limited prospective studies of AI impact on healthcare costs
- Insufficient data on implementation costs and time-to-value
- Lack of standardized economic evaluation frameworks for clinical AI
- Limited assessment of distributional effects (who benefits, who bears costs)
- Inadequate study of impact on healthcare workforce

We recommend HHS support health economics research specifically focused on clinical AI tools, including: - Total cost of ownership studies (implementation, maintenance, training) - Productivity impact assessments - Distributional analyses (effects across payer types, patient populations) - Workforce impact studies

Part II: Response to Proposed Rule on Health Data, Technology, and Interoperability

We strongly support the proposed deregulatory actions and the FHIR-forward vision articulated in this proposed rule. The transition to FHIR-based APIs creates essential infrastructure for AI to advance the Quintuple Aim.

Support for FHIR-Based Approach

The proposed consolidation around FHIR-based APIs directly enables AI deployment and evaluation:

Proposed Change	AI Impact	Quintuple Aim Benefit
FHIR API standardization	Consistent data access reduces integration cost	Cost, Provider Work Life
Removal of redundant criteria	Reduced compliance burden enables innovation focus	Cost
USCDI v3.1 adoption	Standardized data elements improve AI training/evaluation	Population Health, Healthcare Delivery Optimization
Information blocking revisions	AI systems can access data like human users	All dimensions

Modern AI Integration

The rule's reference to emerging standards like Model Context Protocol (MCP) demonstrates awareness of how AI systems interact with data sources. FHIR provides the data layer upon which such protocols can operate. This forward-looking approach positions the United States as a leader in AI-enabled healthcare interoperability.

Comments on Specific Provisions

Certification Criteria Removal (Section III.A)

We support the proposed removal of redundant certification criteria that are no longer necessary given FHIR adoption. The proposal to remove 34 of 60 criteria represents significant burden reduction.

However, we note the following consideration:

Clinical Decision Support (Section III.A.1.b): The proposed revisions maintain core CDS functionality while removing burdensome requirements. We support this approach. For AI-enabled CDS specifically, we recommend guidance on:

- Performance monitoring requirements
- Update/versioning procedures
- User notification of AI involvement

Real World Testing Revisions (Section III.D.3)

We strongly support the proposed changes to Real World Testing (RWT) requirements. The current RWT framework, while well-intentioned, has proven burdensome without proportionate benefit.

The proposed changes align with our experience in federated evaluation:

Current Requirement	Proposed Change	Our Assessment
Annual RWT plans	Descoped	Support: plans often formulaic, limited value
Detailed results reporting	Descoped	Support: focus on meaningful metrics instead
SVAP documentation	Streamlined	Support: accelerates standards adoption

However, we recommend that HHS consider how RWT principles could be adapted for AI-specific evaluation:

- **Federated RWT:** Allow developers to satisfy RWT through participation in federated evaluation networks (like MedPerf) that assess performance across varied real-world settings
- **Continuous monitoring:** Replace annual RWT cycles with **continuous** performance monitoring for AI systems
- **Performance optimization assessment:** Require RWT to include performance stratification across demographic groups

Standards Version Advancement Process (Section III.D.3)

We support streamlining SVAP requirements. Faster standards adoption is particularly important for AI, where capabilities and best practices evolve rapidly.

We recommend: - Expedited SVAP pathways for AI-relevant standards (e.g., FHIR IGs for AI use cases) - Recognition of MLCommons standards (Croissant, AILuminate) as SVAP-eligible - Coordination with FDA on standards for AI-enabled medical devices

Insights Condition Revisions (Section III.D.5)

We support the proposed focus on “use of FHIR in apps through certified health IT” as the primary Insights measure. This metric directly tracks interoperability adoption.

For AI specifically, we recommend HHS consider future Insights measures such as: - Number of AI-enabled applications accessing data through certified APIs - Volume of API calls for AI/ML purposes - Performance metrics for AI systems integrated with certified health IT

USCDI v3.1 Adoption (Section III.B)

We support the adoption of USCDI v3.1 and the iterative process for expanding standardized data elements. For AI applications, we note:

Data Element Category	AI Relevance	Recommendation
Clinical notes	Essential for NLP-based AI	Prioritize in future USCDI versions
Imaging metadata	Critical for radiology AI	Standardize DICOM-FHIR mapping
Social determinants	Enables optimization-focused AI	Expand SDOH elements
Patient-reported outcomes	Supports outcome prediction	Add PRO instruments
Data quality indicators	Enables AI to assess input reliability	Add provenance/quality flags

We recommend that future USCDI development include explicit consideration of AI use cases and engage the AI development community in prioritization.

Information Blocking Revisions (Section IV)

We strongly support the proposed revisions to information blocking definitions and exceptions. These changes directly enable AI deployment while addressing documented misuse.

1. **Clarification that “access” and “use” include automated means (Section IV.A):** This clarification is essential for AI systems that require programmatic access to EHI. Autonomous AI agents and automated data pipelines should be able to access data under the same conditions as human users.

Concrete scenario: *A hospital deploys an AI system for sepsis early warning. The system needs continuous access to vital signs, lab results, and clinical notes. Under current ambiguity, EHR vendors have argued that automated AI access is not covered by information blocking rules. The proposed clarification eliminates this barrier.*

2. **Revision of Infeasibility Exception conditions (Section IV.B):** Removing conditions susceptible to misuse will improve data access for legitimate AI applications.

Concrete scenario: *An AI developer requests API access to evaluate their algorithm across multiple health systems. Under current rules, health systems have cited the “third party seeking modification use” condition to deny access, claiming*

any external evaluation constitutes “modification.” The proposed removal of this condition prevents such misuse.

3. **Manner Exception clarifications (Section IV.C):** Ensuring that data access cannot be impeded by unreasonable contractual terms supports a competitive market for AI tools.
Concrete scenario: *An EHR vendor offers API access but requires AI developers to sign contracts with excessive indemnification, non-compete clauses, or above-market pricing. The proposed clarification that contracts must be “market rate” and cannot be “unconscionable” addresses this barrier.*
4. **TEFCA Manner Exception removal (Section IV.D):** We support this proposal. The information blocking regulations should apply uniformly regardless of the specific exchange mechanism used. This ensures AI systems can access data through any compliant pathway.

Considerations for AI-Enabled Health IT

As HHS resets the Certification Program foundation, we recommend consideration of future requirements specific to AI-enabled health IT. These should be voluntary guidance rather than burdensome certification requirements:

AI Consideration	Recommended Approach	MLCommons Resource
Performance monitoring	Continuous monitoring with drift detection	MedPerf monitoring framework
Version control	Documentation of model versions and changes	GaNDLF versioning
Transparency	Model cards and performance disclosure	AILuminate documentation standards
Performance optimization assessment	Stratified performance reporting	Croissant demographic metadata
Safety evaluation	Testing against edge cases and failure modes	AILuminate safety benchmarks

We recommend HHS develop voluntary guidance that recognizes industry-developed standards (such as those from MLCommons) as acceptable approaches, rather than creating new prescriptive requirements.

Alignment Between Initiatives: A Unified Vision for the Quintuple Aim

These two HHS initiatives are deeply interconnected. Together, they create the foundation for AI to advance all five dimensions of the Quintuple Aim:

AI Adoption Barrier	Deregulatory Solution	Quintuple Aim Impact
Integration complexity	FHIR API standardization	Cost, Provider Work Life
Data access limitations	Information blocking revisions	All dimensions
Evaluation infrastructure	Federated access through standardized APIs	Healthcare Delivery Optimization, Population Health
Innovation barriers	Reduced certification burden	Cost
Performance optimization gaps	Standardized demographics in USCDI	Healthcare Delivery Optimization
Trust deficit	Transparency through recognized standards	Patient Experience

American Leadership

The coordinated approach represented by these initiatives positions the United States as the global leader in AI-enabled healthcare. By establishing: - Open, standardized APIs (FHIR) - Clear rules against information blocking - Reduced regulatory burden on innovation - Support for federated evaluation (preserving privacy while enabling assessment)

The United States can create the model that other nations follow, cementing American competitiveness in the growing global market for healthcare AI.

Conclusion

The MLCommons Medical AI Working Group supports HHS's efforts to accelerate AI adoption in clinical care while modernizing health IT infrastructure. The FHIR-forward approach creates an essential technical foundation for AI deployment, and the focus on reducing unnecessary regulatory burden while preserving safety mechanisms reflects appropriate balance.

AI has the potential to advance all five dimensions of the Quintuple Aim: - **Population**

Health: Early detection, risk stratification, preventive interventions - **Patient Experience:**

Reduced wait times, better coordination, personalized care - **Per Capita Cost:** Administrative automation, reduced waste, optimized resource use - **Provider Work Life:** Documentation

assistance, decision support, reduced burden - **Healthcare Delivery Optimization:**

Performance monitoring across demographics, expanded access

Realizing this potential requires: - Continued coordination across HHS agencies (FDA, CMS, ONC) - Support for open, neutral evaluation infrastructure - Engagement with a broad range of stakeholders including developers, providers, and patients - Recognition of industry-developed standards and best practices - Explicit attention to performance optimization in all AI evaluation and deployment

We offer our expertise and infrastructure to support HHS in achieving these goals. Our MedPerf platform, GaNDLF framework, Croissant metadata standard, and ALLuminate safety benchmarks demonstrate that open, neutral approaches can accelerate AI adoption while maintaining rigorous evaluation standards and optimizing performance across populations.

The alternative to coordinated action is fragmented approaches that slow innovation, create compliance burden, and ultimately delay patient access to beneficial AI tools. With thoughtful implementation of the approaches outlined in these initiatives, HHS can establish American leadership in clinical AI while ensuring safety, effectiveness, and optimized performance for all patients.

Acknowledgments

This response reflects the collective expertise and real-world experience of MLCommons Medical AI Working Group members, including:

- Academic medical centers and community hospitals
- Clinical practitioners across multiple specialties, and other healthcare professionals
- Data scientists and AI researchers
- Patient advocates

We thank all contributors for their insights and commitment to advancing safe, effective, and optimized AI in healthcare. For questions or further discussion, please contact:

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Appendix: Mapping MLCommons Infrastructure to HHS Initiatives

The following table maps specific HHS requirements and goals to MLCommons technical solutions:

HHS Requirement/Goal	MLCommons Solution	Operational Mechanism
Real-world AI evaluation	MedPerf	Federated evaluation across decentralized sites; algorithms travel to data, preserving privacy
Standardized ML workflows	GaNDLF	Zero/low-code framework ensures reproducible AI development pipelines
Dataset interoperability	Croissant	Metadata format enables ML-ready datasets with standardized documentation
AI safety assessment	AILuminate	Benchmark suite evaluates AI risk and reliability across standardized scenarios
Performance optimization assessment	MedPerf + Croissant	Federated evaluation with demographic metadata enables stratified performance analysis
Post-market monitoring	MedPerf monitoring	Continuous performance tracking with drift detection across deployment sites
Transparency / documentation	Model cards + AILuminate	Standardized documentation of AI system characteristics and limitations
FHIR integration	MedPerf FHIR adapters	Reference implementations for AI evaluation on FHIR-accessible data

Example: How MedPerf Addresses Information Blocking Scenarios

Scenario	Current Barrier	How MedPerf Helps
Multi-site AI validation	Data cannot leave institutions	Algorithms travel to data; only aggregate metrics returned
Performance optimization assessment	Demographic data too sensitive to share	Local stratified analysis; only summary statistics aggregated
Continuous monitoring	No infrastructure for distributed surveillance	Federated monitoring dashboard with site-level and aggregate views
Competitive evaluation	Institutions reluctant to share performance data	Neutral third-party orchestration; blinded comparisons

Alignment with Quintuple Aim:

MLCommons Tool	Population Health	Patient Experience	Cost	Provider Work Life	Healthcare Delivery Optimization
MedPerf	Enables validation across populations	Ensures AI works for all patients	Reduces duplicate validation	Streamlines evaluation	Enables stratified assessment
GaNDLF	Reproducible development	Consistent AI behavior	Reduces development cost	Standardized workflows	Reproducible optimization analysis
Croissant	Dataset documentation	Transparency	Reduces data prep cost	Clear data specs	Demographic metadata
AILuminate	Safety assessment	Trust in AI	Reduces safety incidents	Confidence in tools	Systematic error detection

References

- [1] Machine learning is one of the key techniques through which AI systems are built.
- [2] Peter Mattson, et al, "MLPerf: An Industry Standard Benchmark Suite for Machine Learning Performance," IEEE Xplore, 2020. <https://ieeexplore.ieee.org/abstract/document/9001257>
- [3] Karargyris, A., Umeton, R., Sheller, M.J. et al. Federated benchmarking of medical artificial intelligence with MedPerf. Nat Mach Intell 5, 799-810 (2023).
<https://doi.org/10.1038/s42256-023-00652-2>
- [4] Pati, S., Thakur, S.P., Hamamci, I.E. et al. GaNDLF: the generally nuanced deep learning framework for scalable end-to-end clinical workflows. Commun Eng 2, 23 (2023).
<https://doi.org/10.1038/s44172-023-00066-3>
- [5] Akhtar, M., et al. "Croissant: A metadata format for ML-ready datasets." Advances in Neural Information Processing Systems 37 (2024): 82133-82148.
- [6] "The ALLuminate Benchmark," MLCommons, accessed January 2026,
<https://mlcommons.org/ailuminate/>
- [7] LaBella, D., Kassem, H., Edwards, B., Sheller, M., Pati, S., Aristizabal, A., Huang, R., Baid, U., Karargyris, A., Bakas, S., et al. "FeTS 2.0: Federated learning sets benchmark in post-op GBM segmentation." Neuro-Oncology, Volume 27, Supplement 5, November 2025, Page v289.
<https://doi.org/10.1093/neuonc/noaf201.1148>
- [8] Maier-Hein, L., Reinke, A., Godau, P. et al. Metrics reloaded: recommendations for image analysis validation. Nat Methods 21, 195-212 (2024).
<https://doi.org/10.1038/s41592-023-02151-z>
- [9] FDA, "Artificial Intelligence and Machine Learning (AI/ML)-Enabled Medical Devices," updated 2025,
<https://www.fda.gov/medical-devices/software-medical-device-samd/artificial-intelligence-and-machine-learning-aiml-enabled-medical-devices>
- [10] Sahni, N., Stein, G., Zimmel, R., & Cutler, D. M. (2023). The potential impact of artificial intelligence on healthcare spending. The Economics of Artificial Intelligence: Health Care Challenges. University of Chicago Press.
- [11] Vasey, B., et al. (2022). Guidelines and quality criteria for artificial intelligence-based prediction models in healthcare: a scoping review. npj Digital Medicine, 5, Article 2.

List of Acronyms

- ACO: Accountable Care Organization
- AI: Artificial Intelligence
- API: Application Programming Interface
- ASTP: Assistant Secretary for Technology Policy
- CDS: Clinical Decision Support
- CFO: Chief Financial Officer
- CFR: Code of Federal Regulations
- CIO: Chief Information Officer
- CMAO: Chief Medical Analytics Officer
- CMIO: Chief Medical Information Officer
- CMS: Centers for Medicare & Medicaid Services
- CRADA: Cooperative Research and Development Agreement
- CTO: Chief Technology Officer
- CUSUM: Cumulative Sum (control chart)
- DICOM: Digital Imaging and Communications in Medicine
- DSC: Dice Similarity Coefficient
- EHI: Electronic Health Information
- EHR: Electronic Health Record
- EWMA: Exponentially Weighted Moving Average
- FDA: Food and Drug Administration
- FHIR: Fast Healthcare Interoperability Resources
- GBM: Glioblastoma Multiforme
- HHS: Department of Health and Human Services
- HIPAA: Health Insurance Portability and Accountability Act
- IEC: International Electrotechnical Commission
- IEEE: Institute of Electrical and Electronics Engineers
- IG: Implementation Guide
- IT: Information Technology
- KS: Kolmogorov-Smirnov (test)
- MCP: Model Context Protocol
- ML: Machine Learning
- MMD: Maximum Mean Discrepancy
- MSSP: Medicare Shared Savings Program
- NIH: National Institutes of Health
- NLP: Natural Language Processing
- ONC: Office of the National Coordinator for Health Information Technology
- PRO: Patient-Reported Outcomes
- RFI: Request for Information
- RIN: Regulation Identifier Number
- ROI: Return on Investment
- RWT: Real World Testing
- SaaS: Software as a Service
- SDOH: Social Determinants of Health
- SVAP: Standards Version Advancement Process
- TEFCA: Trusted Exchange Framework and Common Agreement
- USCDI: United States Core Data for Interoperability